

David Alexander

davidalx@cs.washington.edu |  david-alexander01 |  ogiorgil

EDUCATION

University of Washington

Seattle, USA

Ph.D. in Computer Science, Database Group. Advised by Magdalena Balazinska.

Sep. 2025 – (expected) Aug. 2030

University of Indonesia

Depok, Indonesia

Bachelor of Computer Science. Cumulative GPA: 3.99/4.0, highest out of more than 400 students.

Sep. 2020 – Dec. 2024

Thesis: Building a Retrieval Module for a RAG System for Data Discovery Queries.

Bangkit Academy led by Google, Tokopedia, Gojek, & Traveloka

Jakarta, Indonesia

Distinguished Cloud Computing Graduate. Google Cloud ACE and PCA certified.

Feb. 2023 – Jul. 2023

PUBLICATIONS

M. I. Balaka, David Alexander, Qiming Wang, Yue Gong, Adila Krisnadhi, and Raul C. Fernandez. **Pneuma: Leveraging LLMs for Tabular Data Representation and Retrieval in an End-to-End System.** *Accepted to SIGMOD 2025.* <https://arxiv.org/abs/2504.09207>.

- Part of an international undergraduate research collaboration program with Professor **Raul C. Fernandez** of the **University of Chicago**.
- Packaged the project and introduced an end-to-end open-source system, accessible on GitHub.

RESEARCH EXPERIENCES

Member of the UW Database group

Sep. 2025 – present

- Exploring novel interfaces between AI agents and databases. My goal is to investigate how database systems can be fundamentally redesigned to better support interaction and reasoning by AI agents.

International Undergraduate Research Collaboration on Data Discovery

Jan. 2024 – Dec. 2024

- Collaborated with Professor **Raul C. Fernandez** of the **University of Chicago** on **Pneuma**.
- Pneuma makes it possible for users to search for relevant tables across databases using natural language, bridging the gap between technical data discovery tools and accessible, intuitive querying. Pneuma also outperforms widely used table search system such as full-text search and state-of-the-art RAG systems in accuracy and resource efficiency.
- Leverage LLMs to represent and retrieve tabular data. Retrieval also uses hybrid search with the BM25 algorithm and HNSW vector index.
- Utilize batch processing on table summarization using binary search to find the optimal batch size, speeding up runtime up to 4×.
- Paper accepted to SIGMOD 2025.
- Supported by the MoECRT Garuda Open Research Program.

WORK EXPERIENCES

Microsoft Research Redmond, Washington

Research Intern

Jun 2026 – Sept 2026

- Mentored by Vivek Narasayya. Worked on problems related to database and AI Agents.

StaffAny Remote, Singapore

Software Engineer

Feb. 2025 – Aug 2025

- Led end-to-end delivery of major product features across web, mobile, and backend platforms, using TypeScript, React, React Native, and Hapi.js. Coordinated multiple engineers and collaborated with the product and design team to align implementation with business goals.
- Delivered key launches including **Social Wall** (company-wide post, like, and comment feature replacing Meta's Workplace Feed) and **Employee Documents**, forming the basis for the upcoming HRAny product line.
- Rearchitected and overhauled core system functionalities used by large enterprise clients with thousands of employees, including **Combined Payment and Bank Files** to streamline multi-payroll processing and **Two Half-Day Leaves in One Day** to support more flexible leave applications.
- Resolved numerous customer-reported issues and production bugs, collaborating with product and support team to quickly address blockers affecting business operations.

Studio S. Jakarta, Indonesia

Frontend Engineer (Freelance)

Jan. 2024 – Feb. 2024

- Developed a customer onboarding feature enabling users to register personalized body measurements using React.

Studio S. *Jakarta, Indonesia*

Backend Engineer (Freelance)

Jul. 2023 – Aug. 2023

- Created a Backend API for a Human Resource Information System software using Express.js + Prisma, deployed on Google Cloud Run.
- Communicated requirements with the client and designed the database schema according to the client's request.

SUR Studios *Jakarta, Indonesia*

Software Engineer Intern

Jun. 2022 – Sep. 2022

- Built a desktop application using C++ and Qt. Dockerized and deployed the application using AWS ECS with autoscaling and load balancing.

TEACHING EXPERIENCES

University of Indonesia *Depok, Indonesia*

Teaching Assistant

Aug. 2021 – Jun. 2023

- Assisted lecturers and undergraduate students in the following courses:
 - * CSCM603154 - Computer Networks (Jan. 2023 – Jun. 2023)
 - * CSGE602040 - Data Structures & Algorithm (Jul. 2022 – Dec. 2022)
 - * CSGE601021 - Programming Foundations 2 with Java (Jan. 2022 – Jun. 2022)
 - * CSCM601150 - Introduction to Digital Systems (Aug. 2021 – Jan. 2022)
- My responsibilities included, but are not limited to:
 - * Preparing assignments
 - * Providing feedback and solutions to the assignments
 - * Giving tutorials and lab sessions to students
 - * Mentoring students on their final project

Peer Tutor for Programming Foundations 1 with Python

Feb. 2021 – Jul. 2021

- Privately gave weekly/by request sessions to tutor 2 students.

ORGANIZATIONAL EXPERIENCES

Orkes Simfoni Universitas Indonesia Mahawaditra *Depok, Indonesia*

Advisory Committee

Jan. 2024 – Present

- Part of a three-person team that provides strategic and technical advice and feedback for the organization's leaders.

President

Jan. 2023 – Dec. 2023

- Led and supervised an 80-person team in various orchestral and organizational projects.

COMPFEST *Depok, Indonesia*

Vice Manager of Competition

Nov. 2021 – Nov. 2022

- Co-led a 60-member team to organize 7 national tech competitions on the topic of AI Innovation Challenge, Business-IT Case, Capture the Flag, Competitive Programming, and Indie Game Development.

Staff of AI Innovation Challenge

Mar. 2021 – Nov. 2021

- Held the first-ever AIC by COMPFEST. Tasked to validate participants' submissions and chosen to be the contact person for any AIC-related inquiries from participants.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, HTML, CSS, SQL, SPARQL

Back End: PydanticAI, Hapi.js, Express.js, Prisma, Node.js, Django REST Framework, Java Spring Boot

Front End: React, React Native, Bootstrap

Developer Tools: Git, Docker, AWS, GCP, PostgreSQL

Selected Coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Advanced Programming, Platform-Based Programming, Database Design, Software Quality Assurance, Image Processing, Computer Networks, Semantic Web

Linguistics: English (Fluent), Indonesian (Native), Bangka Malay (Passive)